ECE501 - Digital Image Processing

Weekly Report: Color Hybrid Images

Week 5

Group 10

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1. Overview of the Past Week

Week 4 was the extension of week 3 as mentioned in earlier report, where we focused more on bringing optimal values of the parameters and adjustment of parameter according to lighting, still facing issue at some extreme exposure of light.

2. Work Completed During the Week

- We reviewed the literature on color hybrid images and frequency decomposition in the RGB color space and the Lab color space.
- Then, we applied low-pass filtering to the first image and high-pass filtering to the second image, channel-wisely.
- Further, transformed images to the Lab color space that offers more luminance-chrominance separation and enhanced perceptual consistency.
- Prepared a small test set of color images with similar characteristics.
- Investigated how visual perception varies with the value of the Gaussian filter sigma.

3. Key Observations

- The colour hybrid images were more informative than the greyscale hybrid images.
- Secondly, lab colour space processing show better alignment between low and high frequency components then rather simply doing RGB and then aligning the two images.

- Here for the color image, finding the optimal version field parameters is difficult because we need to balance the visibility of low and high frequency details for different colors at different distances.
- Selecting optimal Gaussian filter parameters is critical for balancing visibility of low and high frequency details.

4. Challenges Faced

- First we face challenge in the alignment of the colour because at different distances, the colour image show different illusions.
- As mentioned in a key observation, the Gaussian filter parameters where as crucial for the colour images because the contrast of the high and low frequency image change because of the alignment issues.

5. Work Planned for Next Week

- Do experiments with selected subjects and lighting conditions to have better results that vary at different colour intensity.
- Experiment with demorphing techniques to extract low and high frequency images from hybrid images.

6. Conclusion

The group dedicated the week to color hybrid images, achieving perceptual effects. The processing of the lab color space enhanced the clarity of illusions, whereas the difficulties like color artifacts and channel misalignment revealed the areas of improvement.